

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DARES SALAAM REGION

FORM SIX MOCK EXAMINATION

PHYSICS 2

Code: 131/2

Time: 3:00 Hours

Monday, 10th February, 2025 A.M.

INSTRUCTIONS

- This paper consists of six (6) questions.
- Answer FIVE questions.
- Each question carries twenty (20) marks.
- Mathematical tables and non – programmable calculators may be used.
- Write your Index Number on every page of your answer booklet (s) provided.
- The following information may be useful
 - (a) Acceleration due to gravity (g) = 9.8m/s^2
 - (b) Pie, $\pi = 3.14$
 - (c) Molar mass of potassium = 39g
 - (d) Planks constant, (h) = $6.62 \times 10^{-34}\text{Js}$
 - (e) Speed of sound in air $V = 340\text{m/s}$
 - (f) Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12}\text{F/m}$
 - (g) Permeability of free space $\mu_0 = 4\pi \times 10^{-7}\text{H/m}$
 - (h) Atmospheric pressure = $1.01 \times 10^5 \text{ N/m}^2$
 - (i) Young's modulus of aluminum = $7 \times 10^{11} \text{ N/m}^2$
 - (j) Young's modulus of steel = $2 \times 10^{11} \text{ N/m}^2$
 - (k) Density of water (ρ) = 1000kg /m^3
 - (l) $1\text{ev} = 1.6 \times 10^{-19}\text{J}$
 - (m) Rydberg's constant = $1.097 \times 10^7\text{m}^{-1}$
 - (n) Young's modulus of copper = $1.1 \times 10^{11}\text{Pa}$.
 - (o) Mass of electron $m_e = 9.1 \times 10^{-31}\text{kg}$
 - (p) Moral gas constant (R) = $8.3\text{J mol}^{-1} \text{ K}^{-1}$

GEPAM science hub

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1. (a) (i) Water is coming out of a hole made in the wall of tank filled with fresh water. If the size of the tank is increased, will the velocity of efflux of water change?
(02marks)
 - (ii) Water is maintained at a height of 10m in a tank. Calculate the diameter of the circular hole needed at the base of the tank to discharge water at the rate of $26.4\text{m}^3/\text{minute}$
(05marks)
 - (b) (i) Water flows steadily along a horizontal pipe at a rate of $8 \times 10^{-3} \text{ m}^3/\text{s}$. If the area of cross-section of the pipe is $40 \times 10^{-4} \text{ m}^2$, calculate the flow velocity of water
(03marks)
 - (ii) Find the total pressure in the pipe if the static pressure in the horizontal is $3 \times 10^4 \text{ Pa}$
(03marks)
 - (iii) What is the new flow velocity if the total pressure is $3.6 \times 10^4 \text{ Pa}$? (02marks)
 - (c) (i) Show that the quantity $\frac{\rho v D}{\eta}$ (Reynold's number) is dimensionless
(02marks)
 - (ii) Two identical drops of water are falling through air with a steady velocity, V . If the drops combine to form a single drop, what would be the terminal velocity of the single drop?
(03marks)
2. (a) (i) You are provided with two pieces of wooden blocks and a string. Briefly explain how you can demonstrate formation of stationary wave using provided apparatus
(03marks)
 - (ii) Solid can support both longitudinal and transverse waves but only longitudinal waves can propagate in gases. Briefly explain the physics behind (03marks)
 - (b) (i) Sounds or voice seems to sound better than usual when you sing in shower, briefly explain
(02marks)
 - (ii) A fork and a monochord string of length 100cm give 4 beats/s. The string is made shorter without any change of tension, until it is in unison with the fork. If its new length is 99cm, what is the frequency of the fork? (04marks)
 - (c) A wedge-shaped film of air is formed between two parallel sided glass plates by means of straight piece of wire. The two plates are in contact along one edge of the film and the wire is parallel to the thin edge. In such experiment using light of wave length 589nm, the distance between the seventh and one hundred and sixty-seventh dark fringe was 26.3mm and the distance between the junction of glass plates and the wire was 25.6mm. Calculate the angle of wedge and the diameter of the wire. (08marks)
3. (a) (i) The pressure of a gas can be increased by decreasing its volume. Why?
(02marks)
 - (ii) At what temperature is the root-mean-square speed of oxygen molecules equal to the escape velocity from earth's surface?
(03marks)

- (b) (i) Why is soap solution a better cleansing agent than ordinary water?(02marks)
- (ii) If excess pressure is balanced by that due to a column of oil (specific gravity 0.8) 2m high, where $R = 1.0\text{cm}$, find the surface tension of soap solution (04marks)
- (c) (i) Show that elastic energy per unit volume stored in an elastic material is given by $u = \frac{1}{2} E (\text{strain})^2$, where E is the young's modulus of the material (03marks)
- (ii) A composite wire of diameter 1cm consists of copper and steel wires of lengths 2.2m and 2.0m respectively. Total extension of the wire stretched by a force is 1.2 mm. Calculate the force (06marks)
4. (a) With the aid of diagram, describe an experiment to show that excess charge resides only on the outside surface of a hollow conductor (06marks)
- (b) (i) When we introduce a dielectric slab between the two plates of a charged air capacitor its capacitance increases. Explain why? (02marks)
- (ii) An $8 \times 10^{-6}\text{F}$ capacitor is connected in series with $0.5\text{M}\Omega$ resistor across 200V d.c supply. Calculate
- (i) Initial charging current (02marks)
- (ii) The current and potential difference across capacitor 4 second after it is connected to the supply (02marks)
- (c) (i) An infinite number of charges each equal to q are placed along the axis at $x = 1\text{m}$, $x = 2\text{m}$, $x = 4\text{m}$, $x = 8\text{m}$ and so on. Calculate the potential and electric field strength in above set of charges if the consecutive charges have opposite signs (06marks)
- (ii) An electron is situated in uniform field strength of $1.2 \times 10^5\text{V/m}$. Find the force acting on it and its acceleration if it has travelled 20mm from rest (02marks)
5. (a) Briefly explain the following phenomena.
- (i) No force is experienced by a stationary charge in magnetic field (02marks)
- (ii) A light metal disc on top of electromagnet is thrown up as the current is switched on (02marks)
- (iii) The motion of copper plate is damped when it is allowed to oscillate between the two poles of a magnet (02marks)
- (b) (i) Using diagram describe in what way is the behavior of diamagnetic material different from that of paramagnetic material when kept in an external magnetic field. (03marks)
- (ii) A flat circular coil with turns each of area $8 \times 10^{-3}\text{m}^2$ lies on horizontal surface. If the Earth's field has a vertical component of $4 \times 10^{-5}\text{T}$, what will be the size of the pulse of EMF induced in the coil when is flipped over in a time in 0.1 seconds? (04marks)

- (c) A long straight cylindrical wire conductor of radius of 5cm carries a steady current of 2A that is uniformly distributed through the Cross – section of the conductor. Determine the magnetic field.
- (i) At a point 8cm outside the conductor (03marks)
 - (ii) At a point 3cm inside the conductor from center (02marks)
 - (iii) Plot a graph that shown variation of magnetic field with distance from center of coil (02marks)
6. (a) (i) The ionisation energy of hydrogen is 13.6eV. What does it mean?(02marks)
- (ii) The wavelength of the first member of the Balmer series of hydrogen spectrum is $6563\text{\AA}$. Calculate the wavelength of the first member of Lyman series in the same spectrum (04marks)
- (b) Caesium has a work function of 1.9eV. Find
- (i) Its threshold wavelength (02marks)
 - (ii) The maximum energy of the liberated electrons when the metal is illuminated by light of wavelength $4.5 \times 10^{-7}\text{m}$ (02marks)
 - (iii) The stopping potential (02marks)
- (c) (i) What is the significance of binding energy per nucleon? (02marks)
- (ii) A small volume of a solution which contained a radioactive isotope of sodium had an activity of 12,000 disintegrations per minute when it was injected into the blood stream of a patient. After 30 hours the activity of 1.0cm^3 of blood was found to be 0.50 disintegrations per minute. If the half-life of the sodium isotope is taken as 15 hours, estimate the volume of blood in the patient (06marks)

- (a) (i) Velocity of efflux, $v = \sqrt{2gh}$
 Since the velocity of efflux is independent of the area of the hole, it will remain the same. (02)

(ii) Torricelli's theorem:

$$v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 10}$$

$$v = 14 \text{ ms}^{-1}. (01)$$

$$Q = \frac{26.4}{60} \text{ m}^3/\text{s} = 0.44 \text{ m}^3/\text{s} (01)$$

$$Q = AV (01)$$

$$Q = \frac{\pi d^2}{4} \cdot v$$

$$d = \sqrt{\frac{4Q}{\pi v}} = \sqrt{\frac{4(0.44)}{3.14 \times 14}} (01)$$

$$\therefore d = 0.2 \text{ m} = 20 \text{ cm}. (01)$$

(b) (i) $v = \frac{Q}{A} = \frac{8 \times 10^{-3} \text{ m}^3/\text{s}}{40 \times 10^{-2} \text{ m}^2} (01)$

$$\therefore v = 2 \text{ ms}^{-1}. (01)$$

(ii) Total pressure = static pressure + $\frac{1}{2} \rho v^2$ (01)

$$\begin{aligned} \text{Total pressure} &= 3 \times 10^4 + \frac{1}{2} (1000)(2)^2 (01) \\ &= 3.2 \times 10^4 \text{ Pa}. (01) \end{aligned}$$

(ii) $\frac{1}{2}\rho v^2 = \text{Total pressure} - \text{static pressure}$

$$\frac{1}{2}\rho v^2 = (3.6 \times 10^4 - 3.0 \times 10^4) \text{ Pa}$$

$$\frac{1}{2}\rho v^2 = 0.6 \times 10^4$$

$$v = \sqrt{\frac{2(0.6 \times 10^4)}{1000}} = 3.5 \text{ m/s}$$

(c)(i) $\left[\frac{\rho v D}{\eta} \right] = \frac{ML^{-3} \cdot LT^{-1} \cdot L}{ML^{-1} T^{-1}}$

$$= \frac{ML^{-1} T^{-1}}{ML^{-1} T^{-1}}$$

$$\therefore \left[\frac{\rho v D}{\eta} \right] = M^0 L^0 T^0$$

$\therefore$ Reynolds number is dimensionless.

(ii) Volume of bigger drop = Volume of 2 small drops.

$$\frac{4}{3}\pi R^3 = 2\left(\frac{4}{3}\pi r^3\right)$$

$$R = (2)^{1/3} \cdot r$$

$$V_T \propto (\text{radius})^3$$

$$\frac{V_1}{V_2} = \frac{R^3}{r^3} = \left[(2)^{1/3}\right]^3 \cdot r^3$$

$$\therefore \frac{V_1}{V_2} = \frac{2}{1} \quad \therefore V_1 = 2 V_2$$

2

Q (i) Connect two wooden blocks ^{by} provided by string then pluck anywhere between the length of the thread to form stationary wave. The position plucked determines the frequency of vibration of string and hence different pitches are formed. (23)

(ii) Transverse waves travel in form of crests and troughs that involve change in shape of medium. On the other hand longitudinal waves travel in form of compression and rarefaction that involve change in pressure (volume). (23)
Solid have both elasticity of shape and volume hence can support both transverse and longitudinal waves. Furthermore the gases possess only elasticity of volume hence cannot have elasticity of shape, so only longitudinal waves can travel through gases.

2

(b) (i) In shower there is more humidity which decreases density of air as a result velocity of sound waves increases

$$v \propto \frac{1}{\sqrt{\rho}}$$

(02)

(ii) let the fork frequency be f .
When length $l = 99 \text{ cm}$

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$f = \frac{1}{2(99)} \sqrt{\frac{T}{\mu}} \quad \text{--- (i)}$$

When length of string

$$L = 100 \text{ cm}$$

$$f' = f - 4$$

thus,

$$f - 4 = \frac{1}{2(100)} \sqrt{\frac{T}{\mu}} \quad \text{--- (ii)}$$

Dividing eqn (i) by eqn (ii)

$$\frac{f}{f - 4} = \frac{200}{2(99)}$$

$$\therefore f = \frac{400}{4/28} = 400 \text{ Hz}$$

(02)

27
2

(c)

Date given 5/2/20

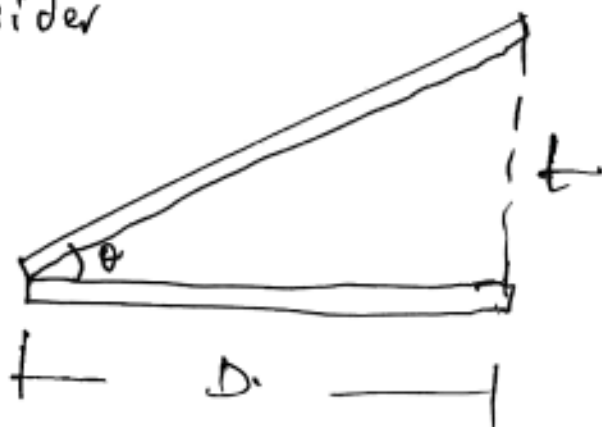
$$\lambda = 589 \text{ nm} = 589 \times 10^{-9} \text{ m}$$

$$\Delta y = y_{107} - y_7 = 26.3 \text{ mm} = 26.3 \times 10^{-3} \text{ m}$$

$$\text{Distance } (D) = 25.6 \text{ mm} = 25.6 \times 10^{-3} \text{ m}$$

$$\text{Diameter } (t) = ?$$

Consider



$$\tan \theta = \frac{t}{D}$$

$$t = D \tan \theta$$

— — — (1)

(62)

Thickness

$$\tan \theta = \frac{t}{y_n}$$

$$\tan \theta = \frac{2t}{2y_n}$$

$$\text{But } 2t = n\lambda$$

$$\tan \theta = \frac{n\lambda}{2y_n}$$

$$y_n = \frac{n\lambda}{2 \tan \theta}$$

(2)

2) (C)

$$y_{167} = \frac{167\lambda}{2 \tan \theta}$$

$$y_7 = \frac{7\lambda}{2 \tan \theta}$$

$$\Delta y = y_{167} - y_7 = \frac{167\lambda}{2 \tan \theta} - \frac{7\lambda}{2 \tan \theta}$$

$$\Delta y = y_{167} - y_7 = \frac{160\lambda}{2 \tan \theta}$$

(02)

$$\tan \theta = \frac{160\lambda}{2 \Delta y} \quad - - - (11)$$

$$\sin \theta = \sin \theta$$

$$\frac{t}{D} = \frac{160\lambda}{2 \Delta y}$$

$$t = \frac{160 D \lambda}{2 \Delta y}$$

$$= \frac{80 \times 25.6 \times 10^{-3} \times 589 \times 10^{-9}}{26.3 \times 10^{-3}}$$

(02)

$$t = 4.587 \times 10^{-5} \text{ m.}$$

3. (a)(i) $P = \frac{\frac{1}{3} N m \bar{c}^2}{V}$; $V = \text{volume.}$

If volume (V) is decreased, the pressure will increase. (02)

(ii) $V_{rms} = \sqrt{\frac{3RT}{m}}$ (001/2)

$T = \frac{m V_{rms}^2}{3R}$ (001/2)

$T = \frac{32 \times 10^{-3} (11.2 \times 10^3)^2}{3(8.3)}$ (01)

$\therefore T = 1.612 \times 10^5 \text{ K.}$ (01)

(b)(i) A cloth has narrow spaces in the form of fine capillaries. The capillary rise is given by

$h = \frac{2\sigma \cos \theta}{\rho g r}$ (001/2)

Addition of soap to water reduces the angle of contact. This will increase the value of $\cos \theta$ and hence h . Thus, the soap solution will rise more in the narrow spaces in the cloth. (001/2)

and clean fabric better than normal water can do alone. (51)

$$(ii) \Delta p = \frac{4\gamma}{R}$$

$$p_2 - p_1 = \frac{4\gamma}{R} \quad (51)$$

Given that

$$p_2 - p_1 = \rho gh$$

$$\Rightarrow \rho gh = \frac{4\gamma}{R} \quad (51)$$

$$\gamma = \frac{\rho gh \cdot R}{4} \quad (51)$$

$$= \frac{800 \times 9.8 \times 2 \times 10^{-3} \times 10^{-2}}{4}$$

$$\therefore \gamma = 3.92 \times 10^{-4} \text{ Nm}. \quad (51)$$

(c) (i) Energy stored, $W = \frac{1}{2} Fe$

$$U = \frac{\text{Energy stored}}{\text{Volume}} \quad (51)$$

$$U = \frac{\frac{1}{2} Fe}{AL}$$

$$U = \frac{1}{2} \left(\frac{F}{A} \right) \left(\frac{e}{L} \right) \quad (51)$$

$$U = \frac{1}{2} (\text{stress}) (\text{strain}) \quad (51)$$

but $E = \frac{\text{stress}}{\text{strain}}$

$\text{stress} = E (\text{strain})$

$U = \frac{1}{2} (E \cdot \text{strain}) \cdot \text{strain}$

$\therefore U = \frac{1}{2} E (\text{strain})^2$

(ii) Young's modulus

$E = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$

$E = \frac{F}{A} \cdot \frac{L}{\Delta L}$

$E_s = \frac{F}{A} \cdot \frac{L_s}{\Delta L_s}$

$E_c = \frac{F}{A} \cdot \frac{L_c}{\Delta L_c}$

$\frac{E_s}{E_c} = \frac{L_c}{L_s} \cdot \frac{\Delta L_s}{\Delta L_c}$

$\frac{\Delta L_s}{\Delta L_c} = \frac{E_c}{E_s} \cdot \frac{L_s}{L_c}$

$\frac{\Delta L_s}{\Delta L_c} = \frac{1.1 \times 10^{11}}{2.0 \times 10^{11}} \times \frac{2}{2.2} = \frac{1}{2}$

$\Delta L_c = 2\Delta L_s$

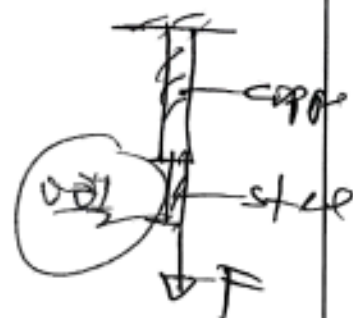
$\Delta L_c + \Delta L_s = 1.2 \text{ mm}$

(Solved)

(Solved)

(Solved)

(Solved)



(Solved)

(Solved)

(Solved)

(Solved)

$$2l_s + l_c = 1.2 \text{ mm}$$

$$3l_s = 1.2 \text{ mm}$$

$$l_s = \frac{1.2 \text{ mm}}{3} = 0.4 \text{ mm}$$

$$l_c = 0.8 \text{ mm} = 8 \times 10^{-4} \text{ m.} \quad \underline{\underline{\text{Sol 2}}}$$

$$A_c = \frac{\pi d_c^2}{4} = \frac{\pi}{4} (1 \times 10^{-2})^2$$

$$= 7.85 \times 10^{-5} \text{ m}^2 \quad \underline{\underline{\text{Sol 2}}}$$

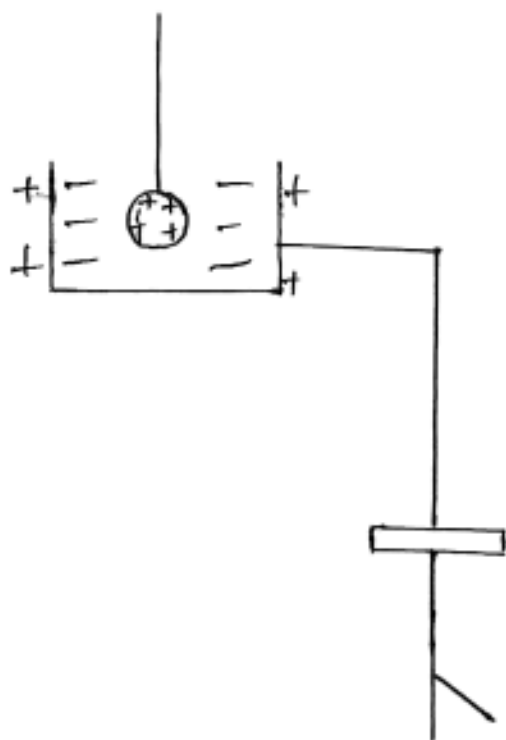
For copper

$$E_c = \frac{F}{A_c} \cdot \frac{L_c}{e_c}$$

$$1.1 \times 10^{11} = \frac{F}{7.85 \times 10^{-5}} \times \frac{2.2}{8 \times 10^{-4}} \quad \text{(5)}$$

$$\therefore \underline{\underline{F = 3.14 \times 10^3 \text{ N.}}} \quad \text{(5)}$$

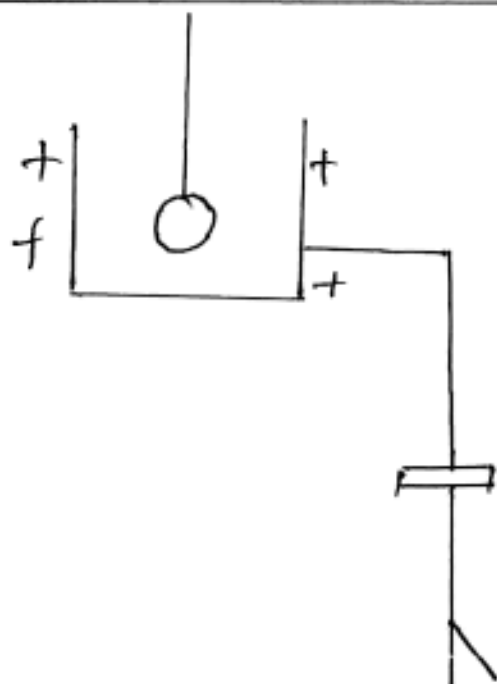
- 4 (a) (i) A metal sphere is tied to an insulating thread and is given a positive charge. The sphere is then lowered into a can connected to the gold leaf electroscope but not allowed to touch the can. (1)
- The leaf diverges. This means that a charge must have been induced outside the can. (1)



- The sphere is then moved ~~but~~ about inside the can and no change in divergence was observed. (1)
- The sphere was then allowed to touch the inside of the can. No change in divergence was observed. (1)

4

(a)



- The sphere was then removed and tested for (1) charge. It was found to have lost all the charge. This means that there was equal and opposite charge inside the can which neutralized the charge on sphere.

CONCLUSION

- When a charged body is enclosed in a (1) hollow conductor, it induces equal and opposite charge on the inside wall, and an equal charge but of the same sign as the charge on the

4 ⑥ ① Introduction of dielectric slab reduces the electric field (E) and hence reduces the potential difference between the plates. Since charge on capacitor plates remain the same, the capacitance of capacitor increases. (02)

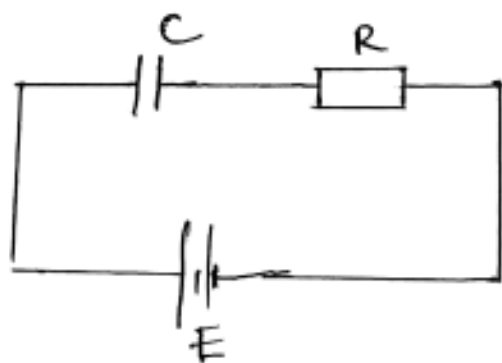
①① Data given

$$C = 2 \times 10^{-6} \text{ F}$$

$$R = 0.5 \times 10^6 \Omega$$

$$E = 200 \text{ V}$$

Consider



① Initial charging current

$$\begin{aligned} I_0 &= \frac{E}{R} \\ &= \frac{200}{0.5 \times 10^6} \end{aligned}$$

$$I_0 = 4 \times 10^{-4} \text{ A}$$

(02)

4 ⑤ (ii) - current after 4 seconds

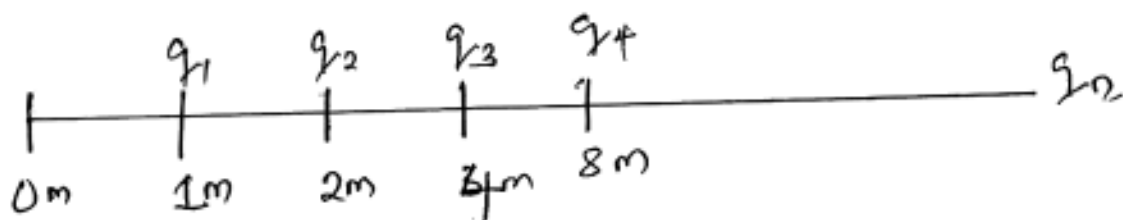
$$I = I_0 e^{-t/CR}$$

$$I = 4 \times 10^{-4} e^{\frac{-4}{(8 \times 10^{-6} \times 0.5 \times 10^6)}}$$

$$I = 1.47 \times 10^{-4} \text{ A}$$

or

⑥ ① Suppose the charges to be in Coulombs and distance in (m) and
Assumption: charges are placed in vacuum



Case I Charges of the same sign
Potential

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right)$$

Taking

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

$$S_\infty = \frac{G_1}{1-r}$$

4

①①

$$S_{\infty} = \frac{1}{1 - \frac{1}{2}} = 2.$$

Then,

$$V = \frac{q}{4\pi\epsilon_0} (2)$$

$$\left(\frac{1}{2}\right)$$

$$V = \frac{q}{2\pi\epsilon_0}$$

Field intensity, E Intensity at $x \rightarrow \infty$,

$$E = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{8^2} + \dots \right)$$

$$= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \right)$$

Taking,

$$1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots$$

$$S_{\infty} = \frac{q}{1 - r}$$

$$\left(\frac{1}{2}\right)$$

$$= \frac{1}{1 - \frac{1}{4}} = \frac{1}{\frac{3}{4}}$$

$$S_{\infty} = \frac{4}{3}$$

4

(c) (i) Then
$$E = \frac{q}{4\pi\epsilon_0} \cdot \frac{4}{3}$$

$$E = \frac{q}{3\pi\epsilon_0}$$

Case II charge of opposite sign

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} \dots \right)$$

Taking

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} \dots$$

$$S_{\infty} = \frac{G}{1-r}$$

$$= \frac{1}{1 - \frac{1}{2}} = \frac{1}{\frac{1}{2}}$$

$$\left(\frac{1}{2} \right)$$

$$S_{\infty} = \frac{2}{1}$$

Then

$$V = \frac{q}{4\pi\epsilon_0} \cdot \frac{2}{1}$$

$$V = \frac{q}{2\pi\epsilon_0}$$

4. ~~Q1~~ Electric field strength

$$E = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r^2} - \frac{1}{2^2} + \frac{1}{4^2} - \frac{1}{8^2} \dots \right)$$

$$E = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{1} - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} \dots \right)$$

Taking

$$1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} \dots$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1}{1 - (-\frac{1}{4})}$$

$$= \frac{1}{1 + \frac{1}{4}} = \left(\frac{4}{5} \right)$$

$$\left(\frac{4}{5} \right)$$

$$S_{\infty} = \frac{4}{5}$$

Then

$$E = \frac{q}{4\pi\epsilon_0} \left(\frac{4}{5} \right)$$

$$E = \frac{q}{5\pi\epsilon_0}$$

4 (c) (ii) Data given

$$E = 1.2 \times 10^5 \text{ V/m}$$

$$d = 20 \text{ mm} = 20 \times 10^{-3} \text{ m}$$

Force on electron

$$F = eE$$

$$F = 1.6 \times 10^{-19} \times 1.2 \times 10^5$$

$$F = 1.92 \times 10^{-14} \text{ N}$$

— Its acceleration (a)

$$F = ma$$

$$a = \frac{F}{m}$$

$$a = \frac{1.92 \times 10^{-14}}{9.1 \times 10^{-31}}$$

$$a = 2.108 \times 10^{16} \text{ m/s}^2$$

(02-marks)

5 (a) (i) From

$$F = BqV$$

Since charge is stationary in magnetic field, $V = 0$

Thus, $F = Bq(0)$. (02)

$$F = 0$$

Therefore, no force is experienced by a stationary charge in magnetic field.

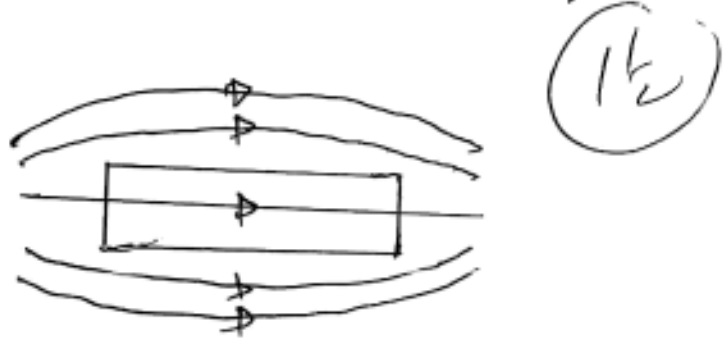
(ii) As the current is switched on, the electromagnet is magnetized thus it attracts the metal disc. So, a change in magnetic flux takes place and an induced emf which according to Lenz's law opposes the cause which produces it hence metal disc is thrown up. (02)

(iii) The cause of damping is the induced emf produced in the copper plates due to change in magnetic flux.

OR
Damping is caused by eddy current generated in copper disc which always opposes the mechanical oscillatory motion. (02)

5

⑥ ① When a diamagnetic material is placed in external magnetic field the field lines are $\perp$ repelled and the lines inside the material is reduced as shown below



(03-Marks)

Whereas, when a paramagnetic material is placed in external magnetic field, the lines get concentrated inside the material and field inside is enhanced as shown below



(1/2)

5) (b) (ii) soln

Data given

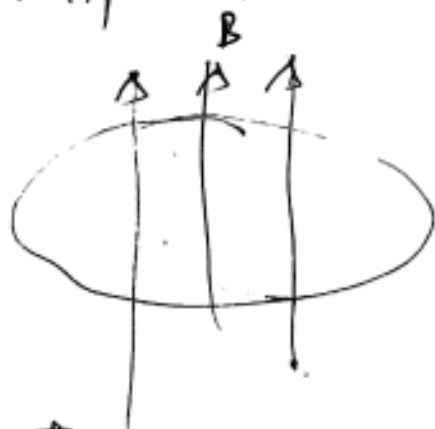
Number of turns (N) = 50 turns

Area (A) = $8 \times 10^{-3} \text{ m}^2$

$B = 4 \times 10^{-5} \text{ T}$

$\Delta t = 0.1 \text{ s}$

Initially ($\theta = 0$)



$$\Phi_i = NBA \cos \theta$$

$$= 50 (4 \times 10^{-5}) (8 \times 10^{-3}) \cos 0$$

$$\Phi_i = 1.6 \times 10^{-5} \text{ Wb}$$

(1)

Finally (Flipped) - $\theta = 180^\circ$



$$\Phi_f = NBA \cos \theta$$

$$= 50 (4 \times 10^{-5}) (8 \times 10^{-3}) \cos 180^\circ$$

$$= -1.6 \times 10^{-5} \text{ Wb}$$

(1)

5) (b) (ii) change in magnetic flux

$$\Delta \Phi = \Phi_f - \Phi_i$$

$$= -1.6 \times 10^{-5} - 1.6 \times 10^{-5}$$

$$\Delta \Phi = -3.2 \times 10^{-5} \text{ Wb.}$$

(01)

Induced Emf

By Faraday's law of induction

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

$$= \frac{-50 \times -3.2 \times 10^{-5}}{0.1}$$

$$= 1.6 \times 10^{-2} \text{ V}$$

(5)

∴ Induced Emf is $1.6 \times 10^{-2} \text{ V}$

6 (c) (i) At 8cm outside the conductor,
from, $B = \frac{\mu_0 I}{2\pi r}$

where, $I = 2A$, $r = 8cm = 0.08m$

$$B = \frac{4\pi \times 10^{-7} \times 2}{2\pi (0.08)}$$

(03)

$$B = 5 \times 10^{-6} T$$

(ii) At 3cm inside the conductor

from, $B = \frac{\mu_0 I r}{2\pi R^2}$

where

$$R = 5cm = 0.05m, \quad r = 3cm = 0.03m$$

$$I = 2A$$

Then

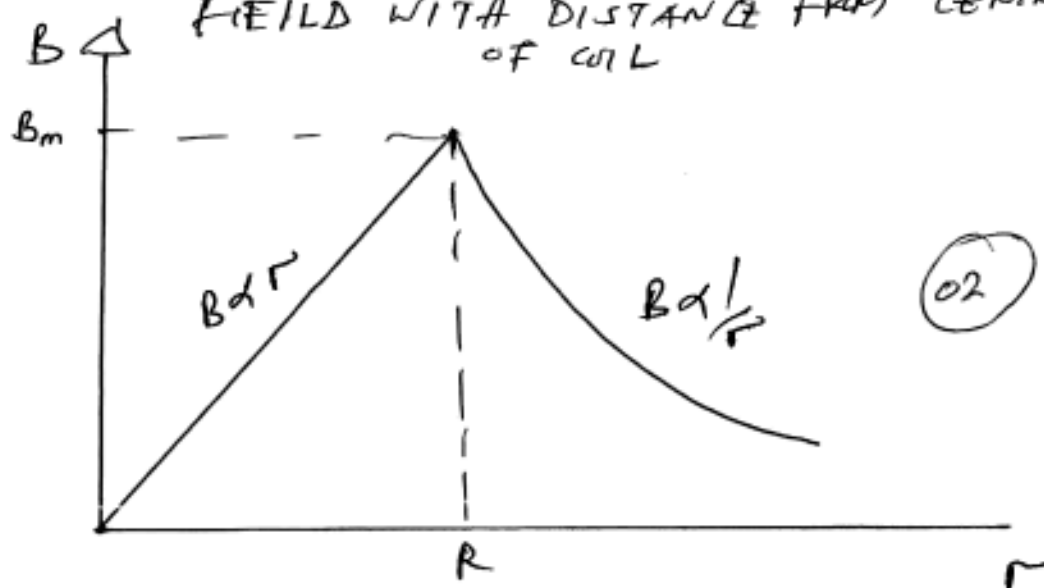
$$B = \frac{4\pi \times 10^{-7} \times 2 \times 0.03}{2\pi (0.05)^2}$$

(02)

$$B = 4.8 \times 10^{-6} T$$

5

C (iii)

GRAPH OF VARIATION OF MAGNETIC
FIELD WITH DISTANCE FROM CENTRE
OF COIL

6. (a) (i) It means that energy required to remove the single electron from the lowest energy level (state) of hydrogen atom to become free electron is 13.6 eV: (02)

$$(ii) \frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad (0.5)$$

For the first member of Balmer series: $n_1 = 2$; $n_2 = 3$ (0.5)

$$\frac{1}{\lambda_{1B}} = R_H \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = \frac{5R_H}{36} \quad (1)$$

For the first member of Lyman series: $n_1 = 1$; $n_2 = 2$ (0.5)

$$\frac{1}{\lambda_{1L}} = R_H \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = \frac{3R_H}{4} \quad (1)$$

Take eqn. (i) $\div$ (ii)

$$\frac{\lambda_{1L}}{\lambda_{1B}} = \frac{5R_H}{36} \times \frac{4}{3R_H} = \frac{20}{108} \quad (0.5)$$

$$\lambda_{1L} = \lambda_{1B} \times \frac{20}{108}$$

$$\lambda_{1L} = 6563 \text{ \AA} \times \frac{20}{108}$$

$$\therefore \lambda_{1L} = 1215 \text{ \AA}.$$

(b) $W_0 = 1.9 \text{ eV}$

$$= 1.9 \times 1.6 \times 10^{-19} \text{ J}$$

$$W_0 = 3.04 \times 10^{-19} \text{ J}.$$

(i) $\lambda_0 = ?$

$$W_0 = \frac{hc}{\lambda_0}$$

$$\lambda_0 = \frac{hc}{W_0} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{3.04 \times 10^{-19}}$$

$$\therefore \lambda_0 = 6.513 \times 10^{-7} \text{ m}.$$

(ii) $\lambda = 4.5 \times 10^{-7} \text{ m}$

$$\frac{hc}{\lambda} = W_0 + KE_{\text{max}}$$

$$KE_{\text{max}} = \frac{hc}{\lambda} - W_0$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4.5 \times 10^{-7}} - 3.04 \times 10^{-19}$$

$$\therefore KE_{\text{max}} = 1.36 \times 10^{-19} \text{ J}.$$

(M) $V_s = ?$

$KE_{\max} = eV_s$ (Q1)

$V_s = \frac{KE_{\max}}{e} = \frac{1.36 \times 10^{-19}}{1.6 \times 10^{-19}}$ (Q2)

$\therefore V_s = 0.85 \text{ V}$ (Q3)

(C) (i) The binding energy per nucleon is an indicator of nuclear stability. The greater the binding per nucleon, the greater is the stability of the nucleus and vice-versa. (Q4)

(ii) $A_0 = 12,000 \text{ dis/min.}$
 $A = ?$

$n = \frac{t}{T_{1/2}} = \frac{30}{15} = 2$ (Q5)

$\frac{A}{A_0} = \frac{1}{2^n}$ (Q6)

$A = \frac{A_0}{2^n} = \frac{12,000 \text{ dis/min}}{2^2}$

$\therefore A = 3000 \text{ dis/min.}$ (Q7)

0.50 dis/min $\xrightarrow{\quad \quad \quad}$ 1.00 dis/min (Q8)
 3000 dis/min $\xrightarrow{\quad \quad \quad}$?

$$? = \text{Volume} = \frac{3000 \text{ g/min} \times 1 \text{ cm}^3}{0.5 \text{ g/min}}$$

$$\therefore \text{Volume of blood} = 6000 \text{ cm}^3$$

